

EXERCISE SHEET 2

This is not the official exercise sheet from the course.

More difficult exercises are denoted by ★ to ★★★

If you can solve it, it is an exercise;
otherwise it's a research problem.

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Exercise 1.

Prove the following statements using a direct proof (sometimes with case distinction)

1. If $n \in \mathbb{N}$, then $n^2 - n$ is even.
2. If $n \in \mathbb{N}$, then $1 + (-1)^n(2n - 1)$ is a multiple of 4.
3. Let $x, y \in \mathbb{Z}$ be odd, then $x^2 + y^2$ is even but not divisible by 4.
4. Let $n \in \mathbb{Z}$ be odd, then $n^2 - 1$ is a multiple of 8.
5. For any $n \in \mathbb{Z}$, n^2 is either a multiple of 3, or one more than a multiple of 3.
6. Every odd integer is the difference of two squares
7. ★ Let $n \in \mathbb{Z}$. If n is not divisible by 3, then $n^2 + 2$ is divisible by 3.
8. ★ Prove that every prime $p > 3$ is of the form $6n \pm 1$ for some integer n .

Exercise 2.

Prove the following statements using an indirect proof.

1. Let $x \in \mathbb{Z}$. If $x^2 - 5x + 6$ is even then x is odd.
2. Let $r \in \mathbb{R}$. If r^2 is irrational, then r is also irrational.
3. Let $x, y \in \mathbb{R}$. If $x + y \geq 20$, then $x \geq 10$ or $y \geq 10$.
4. Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is divisible by 4, then a and b are both even.
5. ★ Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is a perfect square, then at least one of a or b is even.
6. ★ Let $n \in \mathbb{N}$ with $n > 1$. If n is a composite number, then n has a prime factor p such that $p \leq \sqrt{n}$.
7. ★★ Let $k \in \mathbb{N}$. If $2^k + 1$ is prime then either $k = 0$ or $k = 2^n$ for some $n \in \mathbb{N}$.

Exercise 3.

Prove the following statements using a proof by contradiction.

1. The number $\sqrt{3}$ is irrational.
2. There are infinitely many prime numbers.
3. Prove that there are no integers x and y such that $6x + 9y = 100$.
4. Prove that there is no smallest positive rational number.
5. Let q be rational and r irrational. Prove that $q + r$ is irrational.
6. ★ Prove that there are no integers a and b such that $a^2 - 4b = 2$.
7. ★ Let $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, prove that at least one of a or b is divisible by 3.

Exercise 4.

Prove the following statements by mathematical induction

1. For $n \in \mathbb{N}$, $\sum_{i=1}^n (8i - 5) = 4n^2 - n$.
2. For all $n \in \mathbb{N}$, $3^n > n^2$.
3. For all $n \in \mathbb{N}$, $n^3 - n$ is divisible by 3.
4. ★ Let F_n be the n -th Fibonacci number ($F_1 = 1, F_2 = 1, F_{n+2} = F_{n+1} + F_n$).
Prove for all $n \in \mathbb{N}$, $\sum_{k=1}^n F_k = F_{n+2} - 1$.

Exercise 5.

Prove the following statements using the Pigeonhole Principle.

1. Show that in a group of 13 people, at least two were born in the same month.
2. Show that out of 5 random integers, there are two, whose difference is divisible by 4.
3. A round-robin tournament has 10 players. Prove that at any point during the tournament, there are two players who have played the exact same number of matches so far.
4. ★★★ Let $m, n \in \mathbb{N}$. Prove that any sequence of $mn+1$ distinct real numbers contains either a strictly increasing subsequence of length $m+1$ or a strictly decreasing subsequence of length $n+1$.

Exercise 6.

Simplify the following propositional logic formula.

$$F = \neg\left((P \rightarrow Q) \rightarrow (Q \rightarrow P)\right) \vee \left(P \wedge (P \rightarrow Q)\right)$$

Exercise 7.

Simplify the following propositional logic formula.

$$F = \left((P \leftrightarrow Q) \wedge (P \rightarrow \neg Q)\right) \rightarrow \neg P$$

Exercise 8.

Simplify the following formula such that it contains at most one $\neg$, one $\vee$, and one $\wedge$.

$$F = \left((A \rightarrow B) \wedge (B \rightarrow \neg A)\right) \vee \neg\left(A \rightarrow \neg(B \wedge C)\right)$$

Exercise 9.

Simplify the following formula such that it contains at most one $\rightarrow$ operator and no other logical operators,

$$F = \neg\left(\neg(P \wedge Q) \rightarrow (P \rightarrow \neg R)\right) \vee (Q \rightarrow R)$$

Exercise 10.

Simplify the following formula.

$$F = \left((A \wedge B) \rightarrow C\right) \wedge \left((A \wedge \neg C) \rightarrow \neg B\right) \wedge (A \rightarrow B)$$

Exercise 11.

Simplify the following formula such that it contains at most one $\neg$ and at most one $\wedge$.

$$F = \left((A \wedge B) \vee (A \wedge \neg B) \vee (\neg A \wedge B)\right) \rightarrow \neg(A \leftrightarrow B)$$

Exercise 12.

Simplify the following formula to an equivalent expression containing only $\vee$ operators between propositional variables.

$$F = \neg\left((P \rightarrow Q) \wedge (Q \rightarrow R) \wedge (R \rightarrow P)\right) \vee (P \wedge Q \wedge R)$$

Exercise 13.

Simplify the following formula.

$$F = \left((A \wedge B) \rightarrow C\right) \rightarrow \left(A \rightarrow (B \rightarrow C)\right)$$

Exercise 14.

Simplify the following formula such that it contains only $\wedge$ and $\neg$ operators.

$$F = \neg \left((P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow R) \right) \wedge R$$

Exercise 15.

Simplify the following formula.

$$F = \left((A \rightarrow B) \wedge (B \rightarrow C) \right) \wedge \neg(A \rightarrow C)$$